

Birla Institute of Technology and Science Pilani (Rajasthan)
Department of Electrical and Electronics Engineering
EEE F313/ INSTR F313, Analog and Digital VLSI Design
First Semester
Mid-semester Test (Open Book)

[Time: 90 min]

Max. Marks: 75

Date: 31st Oct 2022

General Instructions to the candidate

- Please write your answers legibly and neatly in the answer booklet provided only.
- Symbols, constants, and terminology used have their usual meaning unless specified especially.
- There are a total of **THREE** questions, and all are compulsory. Each carries marks as indicated.

Unless specifically mentioned—

Common data: Use the following common data if not mentioned specifically in the question

Take $V_{DD}=2.5$ V, $V_{TON}=0.4$ V, $V_{TOP} = -0.4$ V, $R_{nmos} = 13$ kohm/ unit (W/L), $R_{pmos} = 31$ kohm/ unit (W/L), $L_{min}=250$ nm, $\lambda_n=0.1$ V⁻¹, $\lambda_p = -0.1$ V⁻¹, $\mu_n C_{ox} = 120$ μ A/ V², $\mu_p C_{ox} = 40$ μ A/ V²

• Use square law current equation unless specifically mentioned. Use lambda-based design rules unless specially mentioned.

• Neglect body effect, and channel length modulation if not mentioned specifically.

• The body of all PMOS is tied to V_{DD} while all NMOS is tied to the ground unless specially mentioned.

• For standard CMOS inverter, $t_{phl}=t_{plh}$, $g=1$, $p=1$. Given $\tau_{250nm} = 0.5$ ns

Answer all the subparts of a question together in sequence.

Q1. A) A three-finger transistor with (W/L) = 20 is given in Fig. 1

Given: $\lambda = 0.25$ μ m, $C_j = 2.4 \times 10^{-4}$ pF/ μ m²,
 $C_{jsw} = 2.0 \times 10^{-4}$ pF/ μ m

- Mark Drain and Source junctions in Fig. 1
- Now, calculate the Drain junction to bulk capacitance for Fig. 1. Ignore the overlap capacitance.
- If this layout is being fabricated using CMOS processing steps, which kind of lithography would be preferred here: photolithography, E-beam lithography, ion-beam lithography, and soft lithography? Justify your answer.

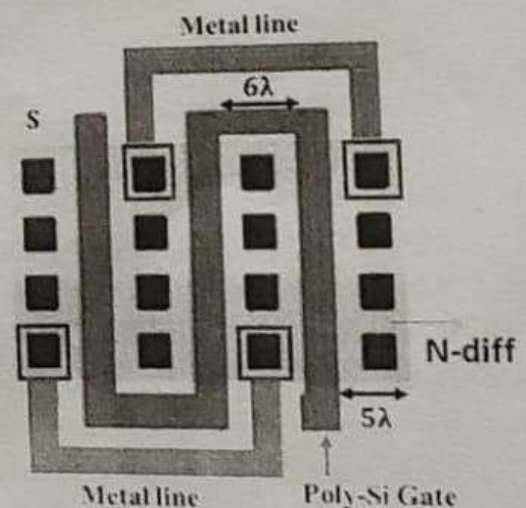


Fig. 1

- B) A digital system formed with $0.25\ \mu\text{m}$ technology has the following specifications: $f_{\text{clock}} = 0.1\ \text{GHz}$, $V_{\text{DD}} = 2.2\ \text{V}$, Power consumption = $5\ \text{W}$. Use long-channel current equation.
- If constant voltage scaling is used to scale down the technology to $90\ \text{nm}$, calculate the new f_{clock} , V_{DD} , and Power consumption.
 - If V_{DD} is chosen to be $1\ \text{V}$ when the processor is scaled down to $90\ \text{nm}$ technology, find the f_{clock} and Power consumption.
 - What should be the supply voltage if it is desired to maintain the power consumption constant while scaling down to $90\ \text{nm}$ technology? How will the maximum clock frequency change in such a case?

[25 marks]

Q2. Consider the circuit shown in Fig. 2 is designed in a $250\ \text{nm}$ technology node.

$C_L = 400\ C_{g,\text{min}}$ and $C_A = 4\ C_{g,\text{min}}$. Use common data given.

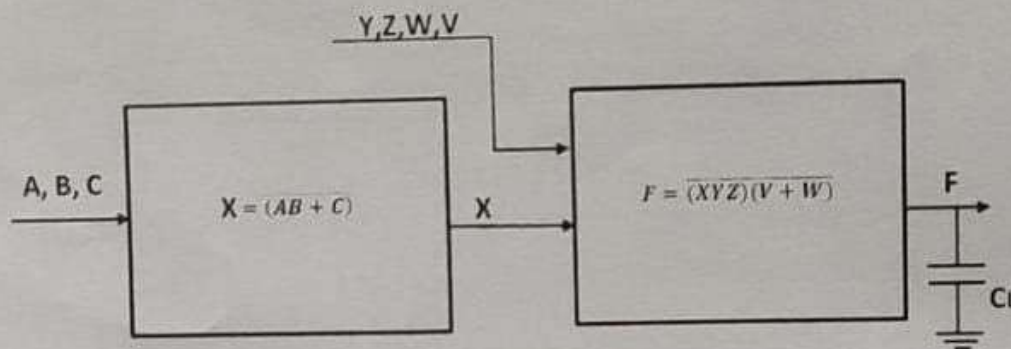


Fig. 2

- Sketch and label the schematic of Fig. 2 using full static CMOS logic with minimum parasitic capacitances at output node/s.
- Sketch and label the worst case equivalent inverter representation of Fig. 2. Hence, write the expression of $(W/L)_{\text{equivalent}}$ of pull down and pull up network.
- Determine the least path delay from signal A to signal F. Hence determine the maximum frequency of operation at which the circuit of Fig. 2 can operate.
- Determine the dynamic power consumption of Fig. 2. Assume $C_{g,\text{min}} = 10\ \text{fF}$.
- Now the circuit of Fig. 2 is designed in $180\ \text{nm}$ technology using constant electric field scaling. Again, determine the dynamic power consumption of Fig. 2.

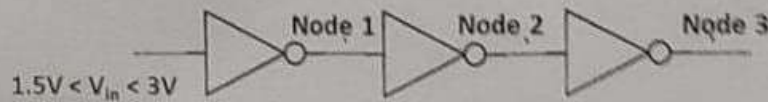
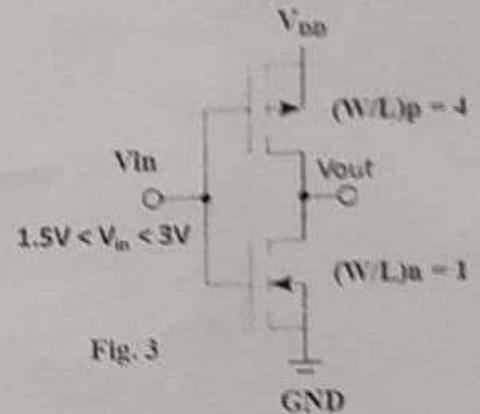
[25 marks]

Q3. Consider a CMOS inverter as shown in Fig. 3. The input (V_{in}) to the inverter is-

$1.5V < V_{in} < 3V$. (Use long channel approximation)

Take : $V_{DD} = 5V$, $k_n' = 600\mu A/V^2$, $k_p' = 240\mu A/V^2$, $V_{thn} = 1V$, $V_{thp} = -1V$.

- j. Calculate V_{OH} and V_{OL} .
- ii. Calculate the inverter's power consumption when the input transitions from 1.5V to 3V. The inverter is now used to drive a load capacitance of 10pF and is operated at a frequency of 10 MHz. Determine the total power consumption.
- iii. Due to operation in a noisy environment, V_{OH} gets degraded by $\pm 20\%$ V_{DD} and V_{OL} by $\pm 10\%$ V_{DD} . What should be the switching threshold, to ensure that $NM_{H1} = NM_{L1}$? Now determine the NMOS to PMOS ratio to achieve this switching threshold.
- iv. The inverter of the part (iii) (with degraded logic levels) is now used in a circuit as shown in Fig. 4. Additional noise of $\pm 2V$ (due to cross talk) gets coupled at Node 1. Determine if the function at Node 2 gets corrupted or not. Justify your answer.



[25 marks]

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